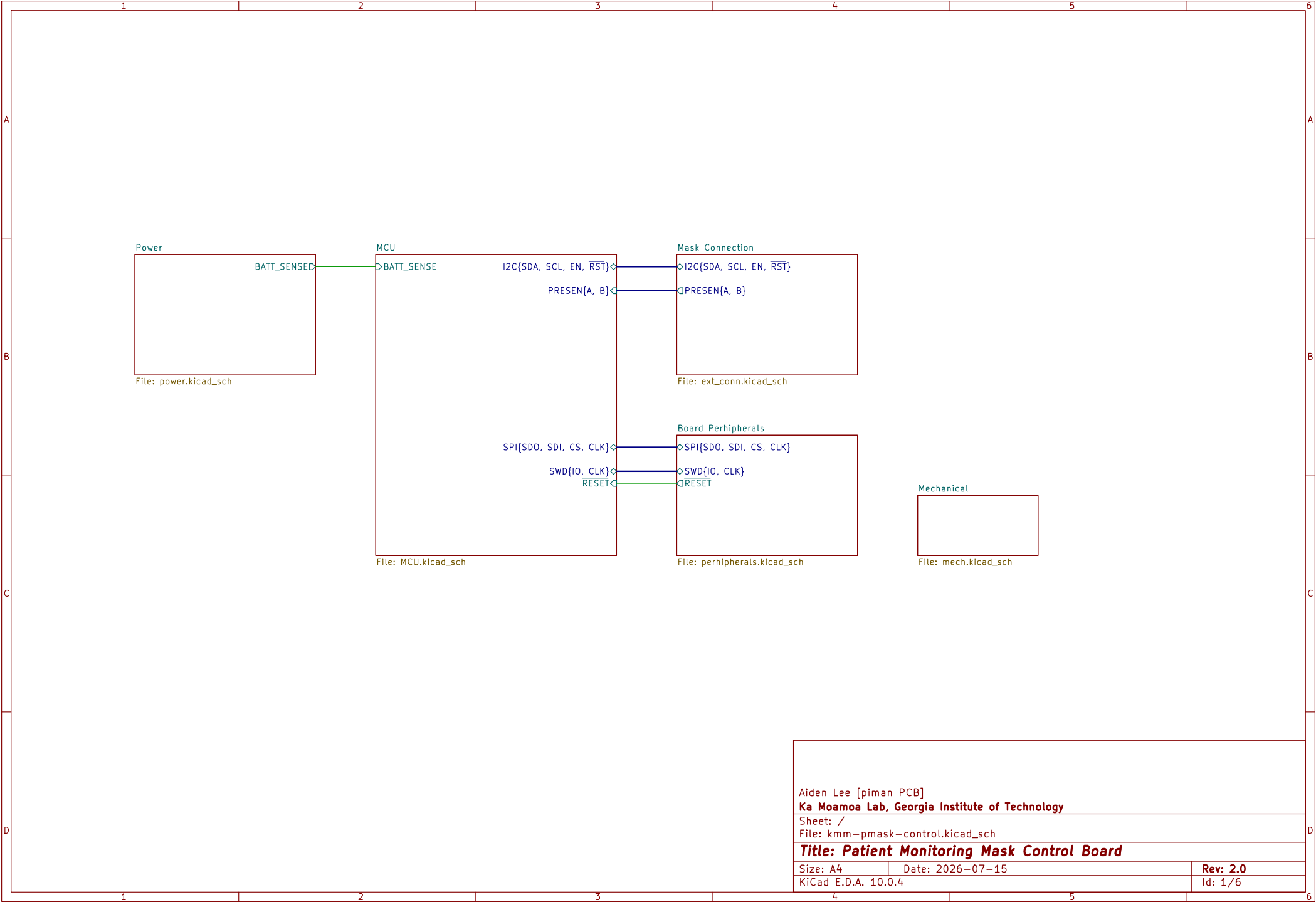




Battery Input Terminals

S2B-PH-K-5

1 BATT_RAW_P
2 BATT_RAW_N

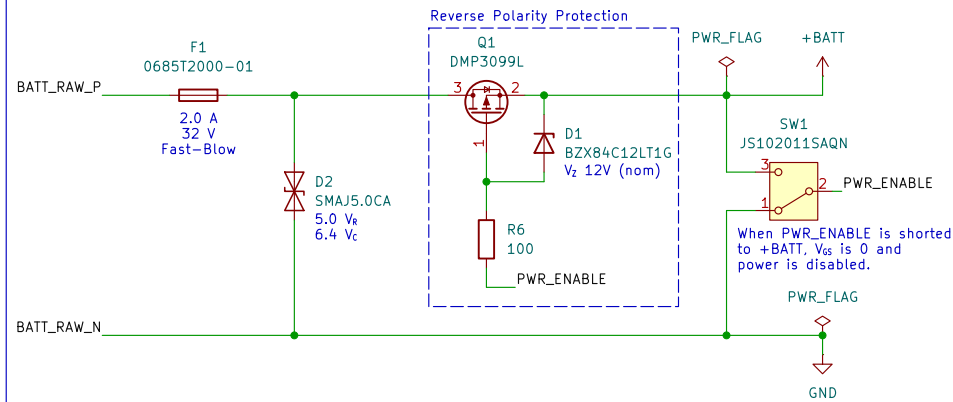
M20-9750246

1 BATT_RAW_N
2 BATT_RAW_P

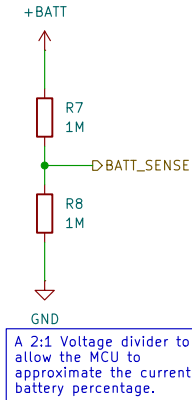
Expected battery chemistry:
1S LiPo, unknown capacity

Customer requested multiple battery
connection points, including a 2.54 mm
pin header.
Do not connect both at the same time.

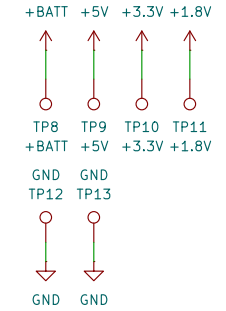
Battery Input Protection



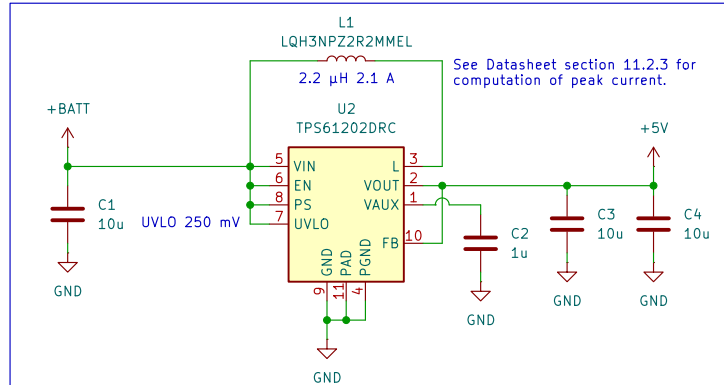
Battery Voltage Probe



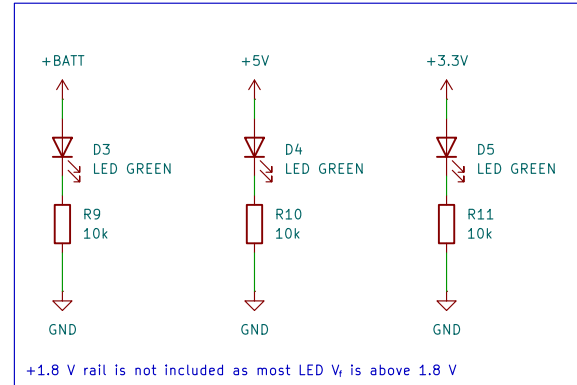
Test Points



800 mA 5V Boost converter



Power rail status indicators



Recommended Supply Current per Power Rail, Slave Board

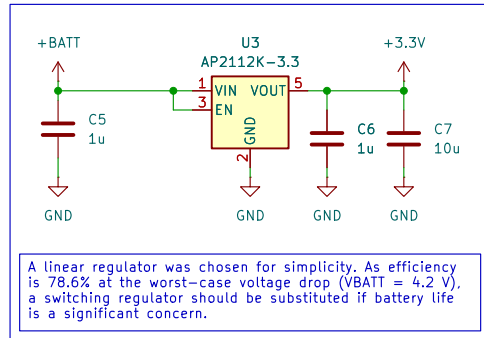
Condition	1.8 V	3.3 V	5.0 V
Recommended Minimum	5 mA	350 mA	800 mA
Maximum	10 mA	400 mA	1000 mA

Maximum Current Consumption per Power Rail, This Board

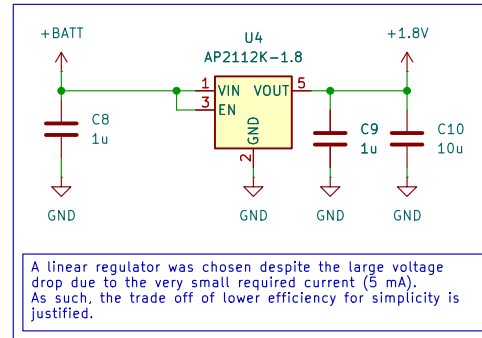
Part #	Quantity	1.8 V	3.3 V	5.0 V
MDBT50Qx	1	-	<100 mA	-

It is very difficult to estimate the actual power consumption of a board until a prototype is built. As such, this revision is built to support the maximum requirements of all components. In future revisions, it is perfectly fine to reduce current supply capability to better align with our usage patterns.

600 mA 3.3V LDO



600 mA 1.8V LDO



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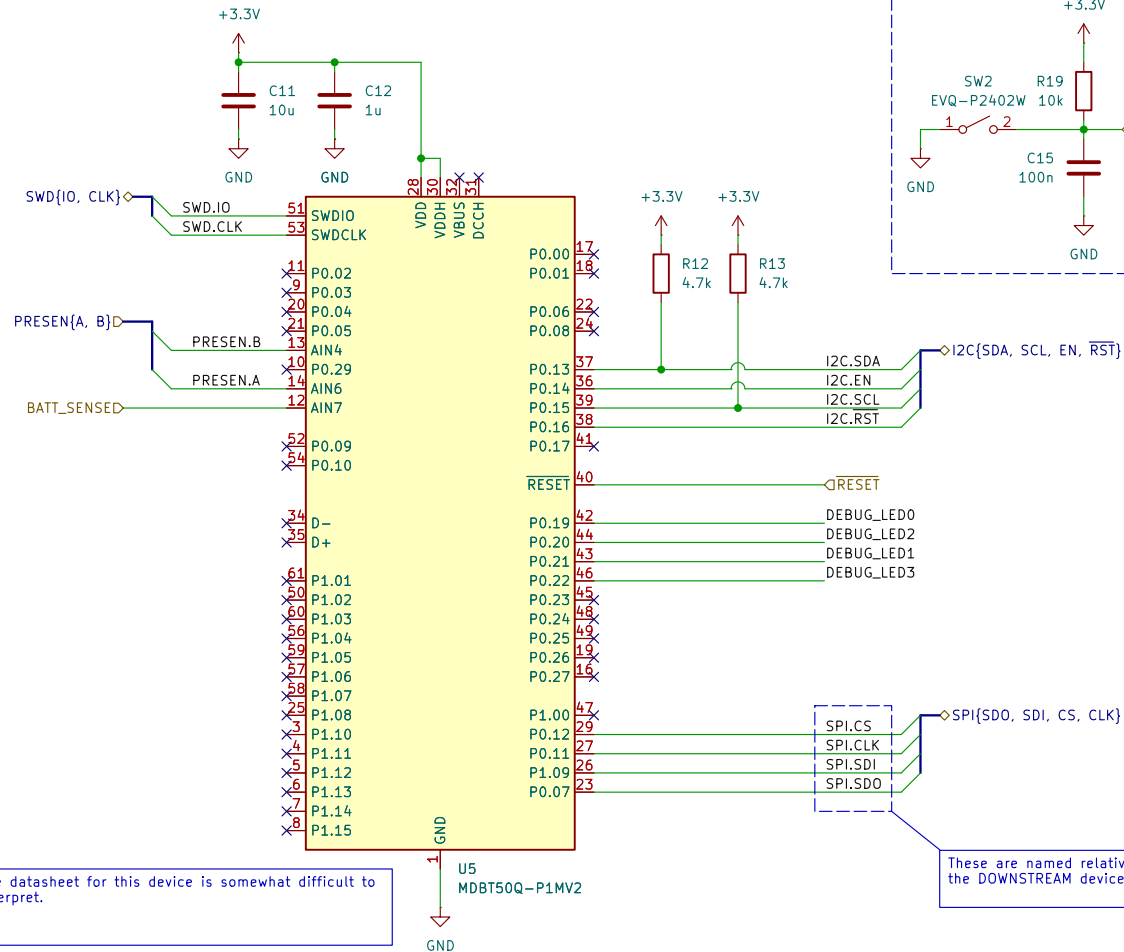
Sheet: /Power/
File: power.kicad_sch

Title: Power Management

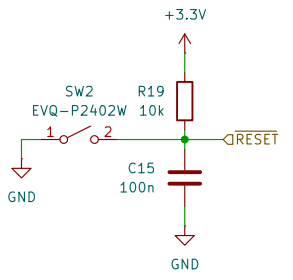
Size: A4 Date: 2026-07-15
KiCad E.D.A. 10.0.4

Rev: 2.0
Id: 2/6

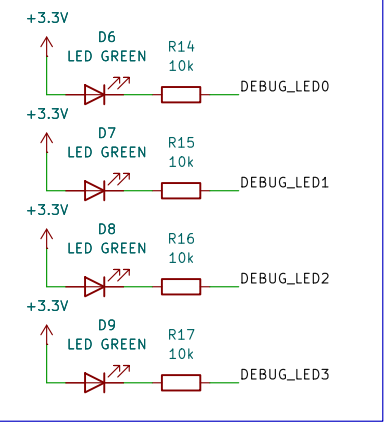
nRF52840 SIP



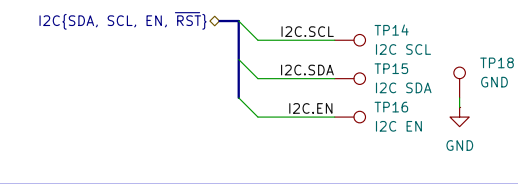
Reset Button



General-Purpose Debug LEDs



Test Points, I2C



These are named relative to the DOWNSTREAM device!

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Sheet: /MCU/

File: MCU.kicad_sch

Title: MCU

Size: A4

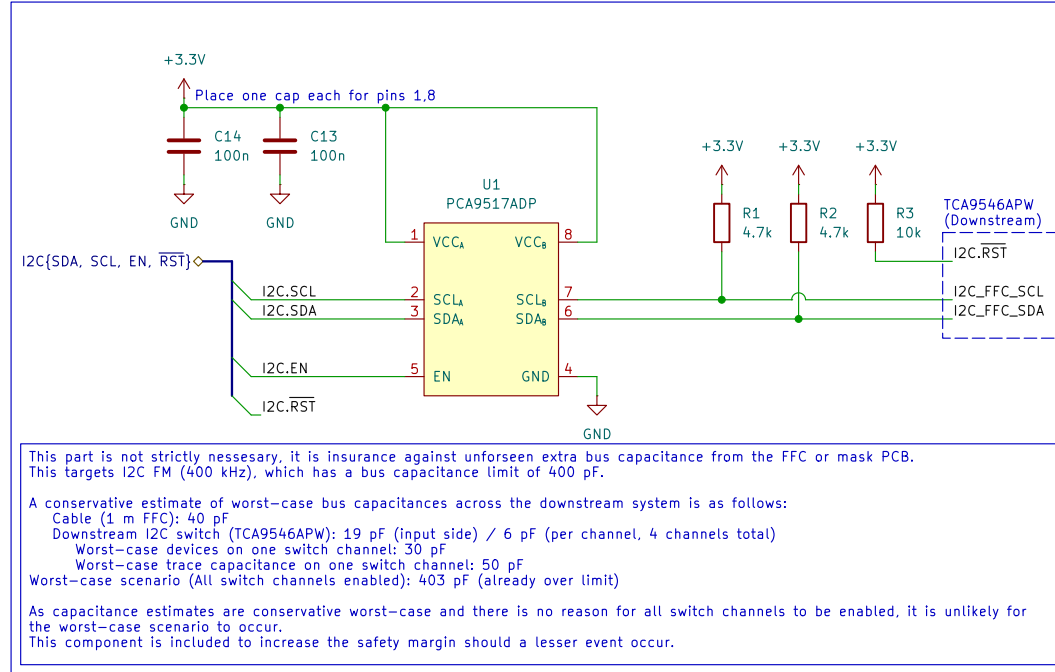
Date: 2026-07-15

Rev: 2.0

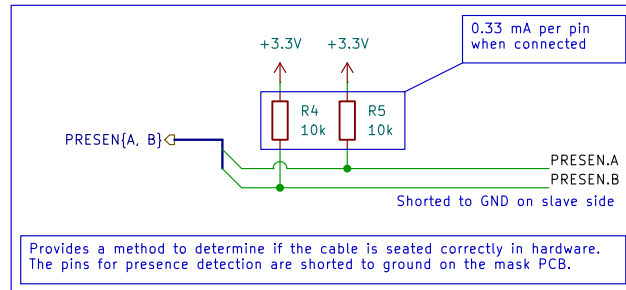
KiCad E.D.A. 10.0.4

Id: 3/6

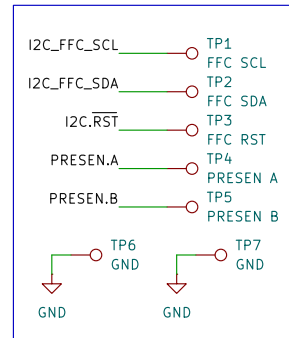
I2C Bus Repeater (400 kHz)



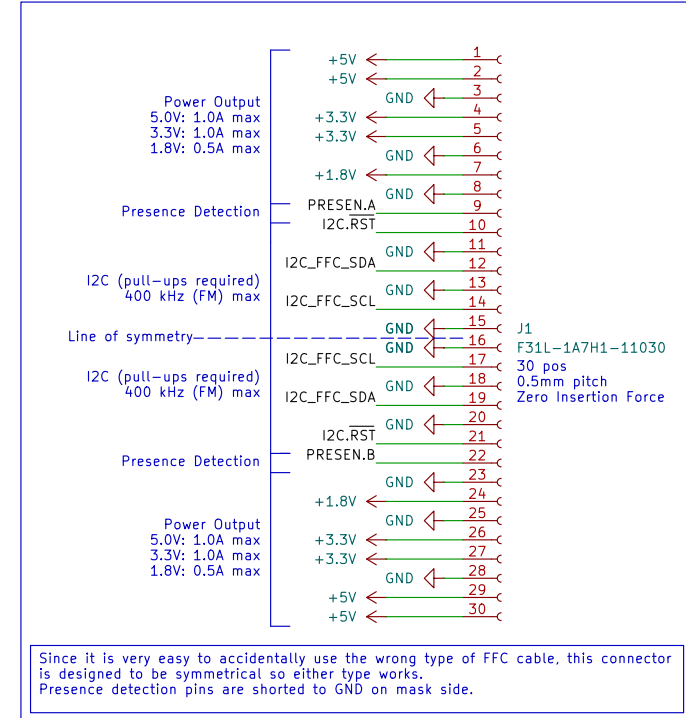
Cable Presence Detection



Test Points



External FFC Connection



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Sheet: /Mask Connection/

File: ext_conn.kicad_sch

Title: External FFC Connection to Mask

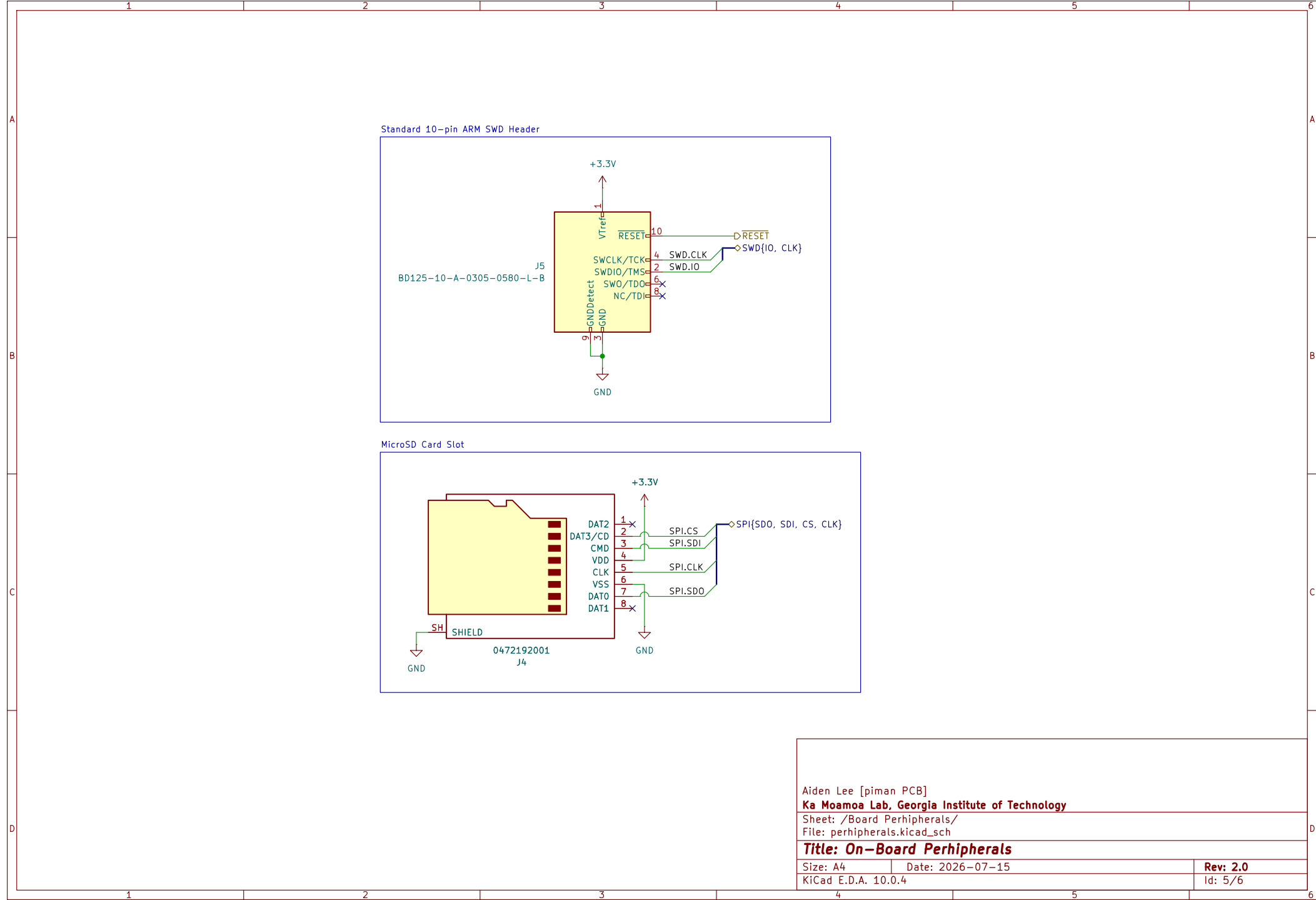
Size: A4

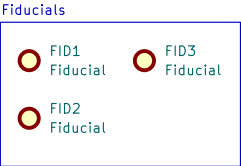
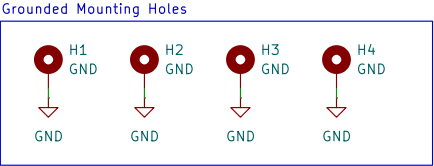
Date: 2026-07-15

Rev: 2.0

KiCad E.D.A. 10.0.4

Id: 4/6





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Sheet: /Mechanical/
File: mech.kicad_sch

Title:

Size: A4	Date: 2026-07-15	Rev: 2.0
KiCad E.D.A. 10.0.4		Id: 6/6